

Data Converters

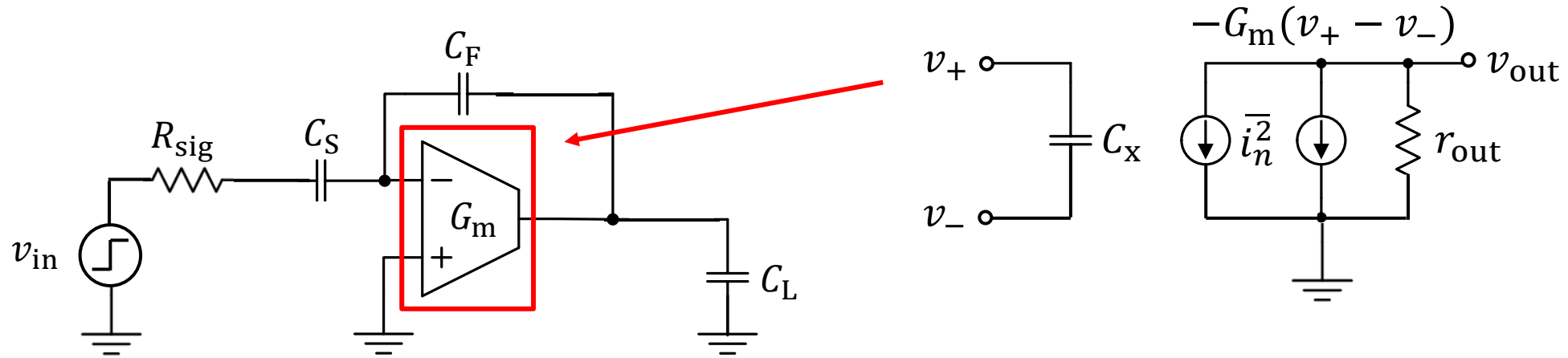
Exercise IV

Summer Term 2026

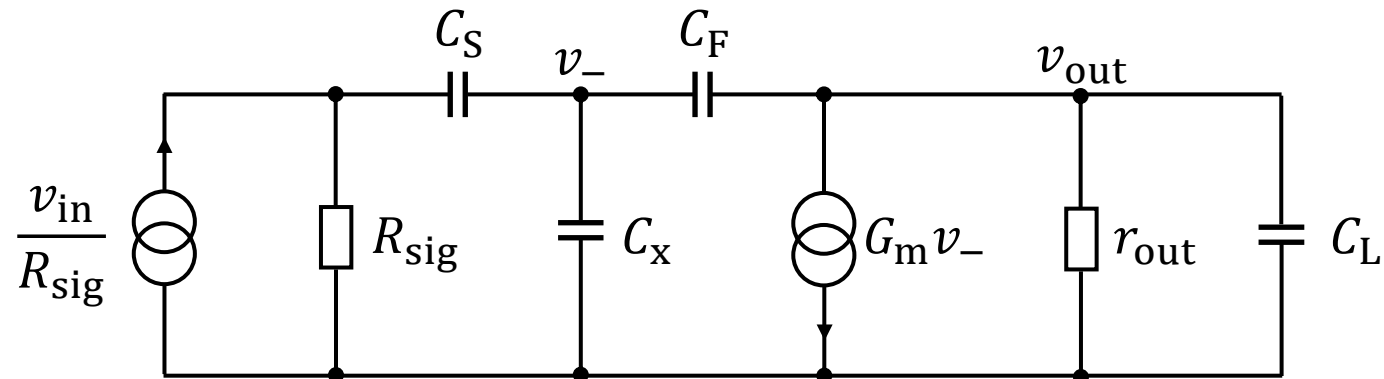
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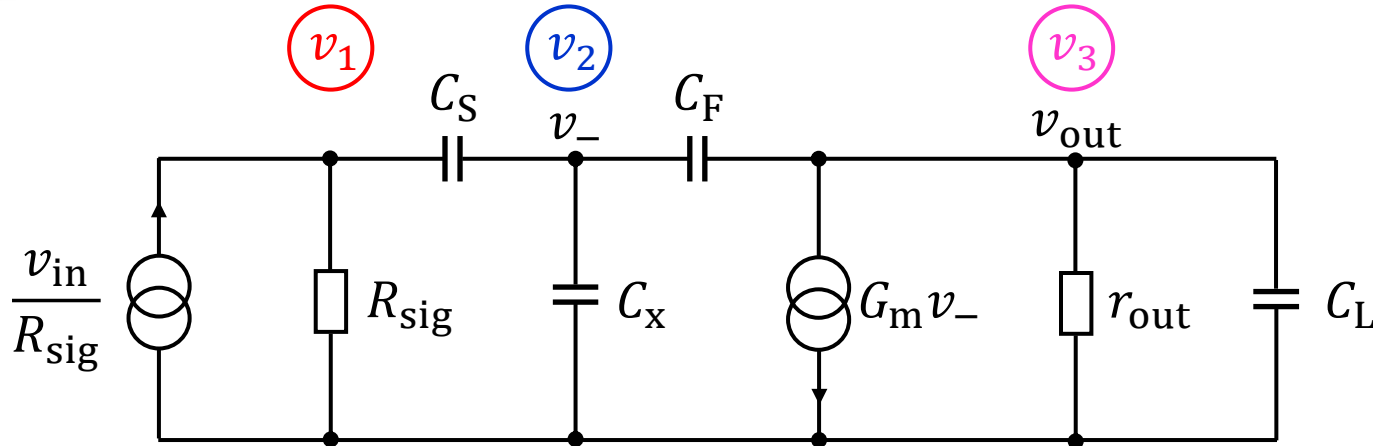
TI.a – Small-signal Equivalent Circuit



small-signal equivalent circuit



TI.b – Nodal Analysis



incoming currents → negative currents
outgoing currents → positive currents

node 1: $G_{\text{sig}}v_1 + sC_S(v_1 - v_2) = G_{\text{sig}}v_{\text{in}}$

node 2: $-sC_S(v_1 - v_2) + sC_Xv_2 + sC_F(v_2 - v_3) = 0$

node 3: $-sC_F(v_2 - v_3) + G_mv_2 + g_{\text{out}}v_3 + sC_Lv_3 = 0$

$$\begin{pmatrix} G_{\text{sig}} + sC_S & -sC_S & 0 \\ -sC_S & s(C_S + C_X + C_F) & -sC_F \\ 0 & -sC_F + G_m & g_{\text{out}} + s(C_F + C_L) \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} G_{\text{sig}}v_{\text{in}} \\ 0 \\ 0 \end{pmatrix}$$

TI.b – Nodal Analysis

$$\begin{pmatrix} G_{\text{sig}} + sC_S & -sC_S & 0 \\ -sC_S & s(C_S + C_X + C_F) & -sC_F \\ 0 & -sC_F + G_m & g_{\text{out}} + s(C_F + C_L) \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} G_{\text{sig}} v_{\text{in}} \\ 0 \\ 0 \end{pmatrix}$$

Solve the system by means of Mathematica or Maple to get

$$TF(s) = \frac{v_{\text{out}}}{v_{\text{in}}} = \frac{v_3}{v_1} = \frac{-C_S(G_m - sC_F)}{(C_F + C_S + C_X)g_{\text{out}} + C_F G_m + s(C_F C_L + C_F C_S + C_L C_S + C_F C_X + C_L C_X)}$$

TI.c – Simplification for $g_{\text{out}} \ll G_{\text{m}}$

$$\begin{aligned} TF(s) = \frac{v_{\text{out}}}{v_{\text{in}}} = \frac{v_3}{v_1} &= \frac{-C_S(G_{\text{m}} - sC_F)}{(C_F + C_S + C_X)g_{\text{out}} + C_F G_{\text{m}} + s(C_F C_L + C_F C_S + C_L C_S + C_F C_X + C_L C_X)} \\ &\approx \frac{-C_S(G_{\text{m}} - sC_F)}{C_F G_{\text{m}} + s(C_F C_L + C_F C_S + C_L C_S + C_F C_X + C_L C_X)} \\ &= -\frac{C_S}{C_F} \frac{1 - s/(G_{\text{m}}/C_F)}{1 + s/\{G_{\text{m}}/(C_L + C_S + C_X + (C_X + C_S)C_L/C_F)\}} \end{aligned}$$

TI.d – DC Gain

$$TF(s) = \frac{v_{\text{out}}}{v_{\text{in}}} = \frac{v_3}{v_1} \approx -\frac{C_S}{C_F} \frac{1 - s/(G_m/C_F)}{1 + s/\{G_m/(C_L + C_S + C_X + (C_X + C_S)C_L/C_F)\}}$$

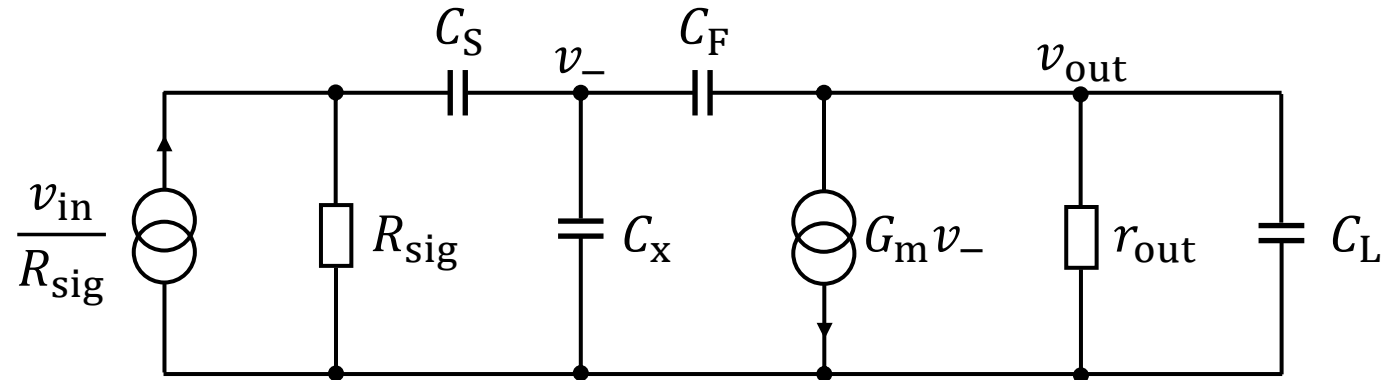
$$A_{\text{DC}} = TF(s = 0) = -\frac{C_S}{C_F}$$

TI.e – Determination Zero

$$TF(s) = \frac{v_{\text{out}}}{v_{\text{in}}} = \frac{v_3}{v_1} \approx -\frac{C_S}{C_F} \frac{1 - s/(G_m/C_F)}{1 + s/\{G_m/(C_L + C_S + C_X + (C_X + C_S)C_L/C_F)\}}$$

$$z = \frac{G_m}{C_F}$$

TI.f – Intuitive Determination Pole



C_F is connected between input and output of an amplifier
 → apply Miller-theorem
 → C_F parallel C_L at the output node

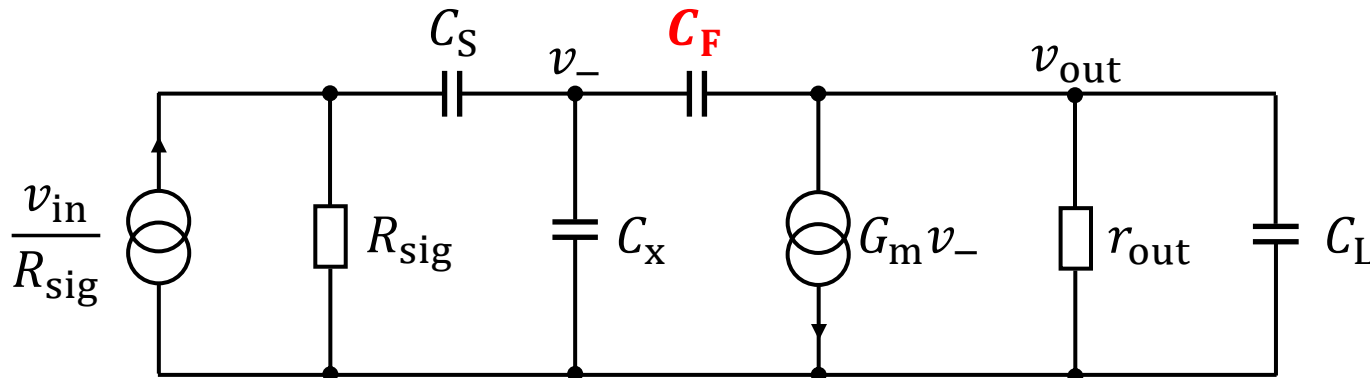
$$p_{\text{est}} \approx -\frac{1}{r_{\text{out}}(C_F + C_L)}$$

TI.g – Determination Pole

$$TF(s) = \frac{v_{out}}{v_{in}} = \frac{v_3}{v_1} \approx -\frac{C_S}{C_F} \frac{1 - s/(G_m/C_F)}{1 + s/\{G_m/(C_L + C_S + C_X + (C_X + C_S)C_L/C_F)\}}$$

$$p = -\frac{G_m}{C_L + C_S + C_X + \frac{(C_X + C_S)C_L}{C_F}} \quad \leftrightarrow \quad p_{est} \approx -\frac{1}{r_{out}(C_F + C_L)}$$

Pole determined by G_m and not r_{out} !



**C_F shorts v_- and v_{out}
 $\rightarrow 1/G_m$ parallel r_{out}
 $\rightarrow 1/G_m$ determines overall resistance
 (like $1/g_m$ parallel r_{ds} in a diode-configured transistor)**

TI.g – Time-Domain Step Response

$$TF(s) = \frac{v_{\text{out}}}{v_{\text{in}}} = \frac{v_3}{v_1} \approx -\frac{C_S}{C_F} \frac{1 - s/(G_m/C_F)}{1 + s/\{G_m/(C_L + C_S + C_X + (C_X + C_S)C_L/C_F)\}}$$

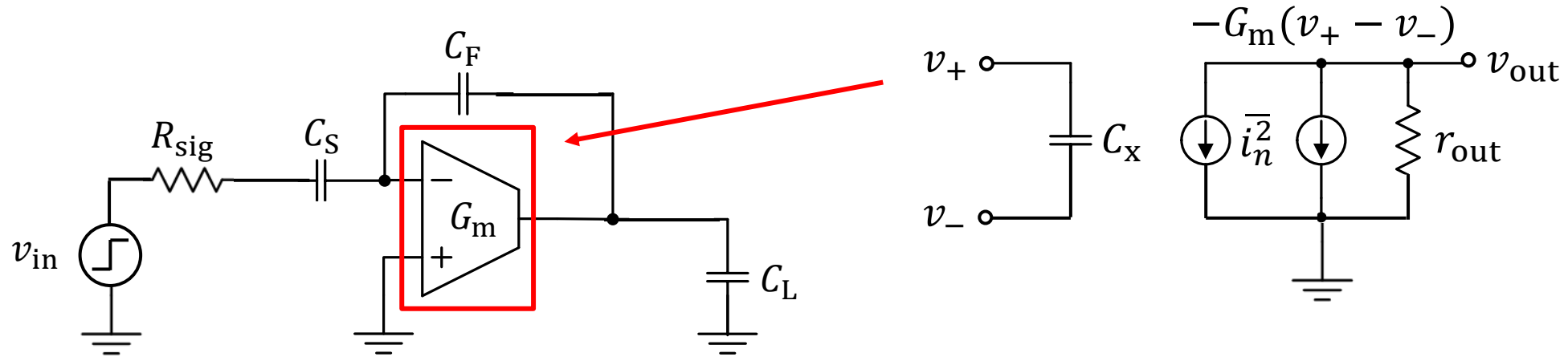
Neglecting the zero and knowing that

$$A_{\text{DC}} = TF(s = 0) = -\frac{C_S}{C_F} \qquad p = -\frac{G_m}{C_L + C_S + C_X + \frac{(C_X + C_S)C_L}{C_F}}$$

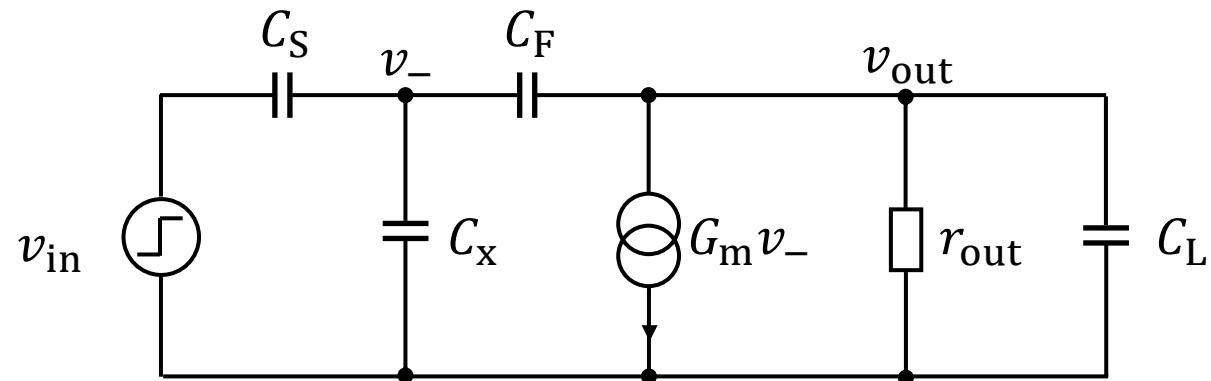
the time-domain step response results in

$$v_{\text{out}}(t) = -\frac{C_S}{C_F} (1 - e^{-t \cdot p}) v_{\text{step}}$$

TII.b – Small-signal Equivalent Circuit



small-signal equivalent circuit



TII.c – Determination Loop Gain $L(s)$

step I: deactivate the signal source

step II: break the feedback loop
(typically @ amplifier input)

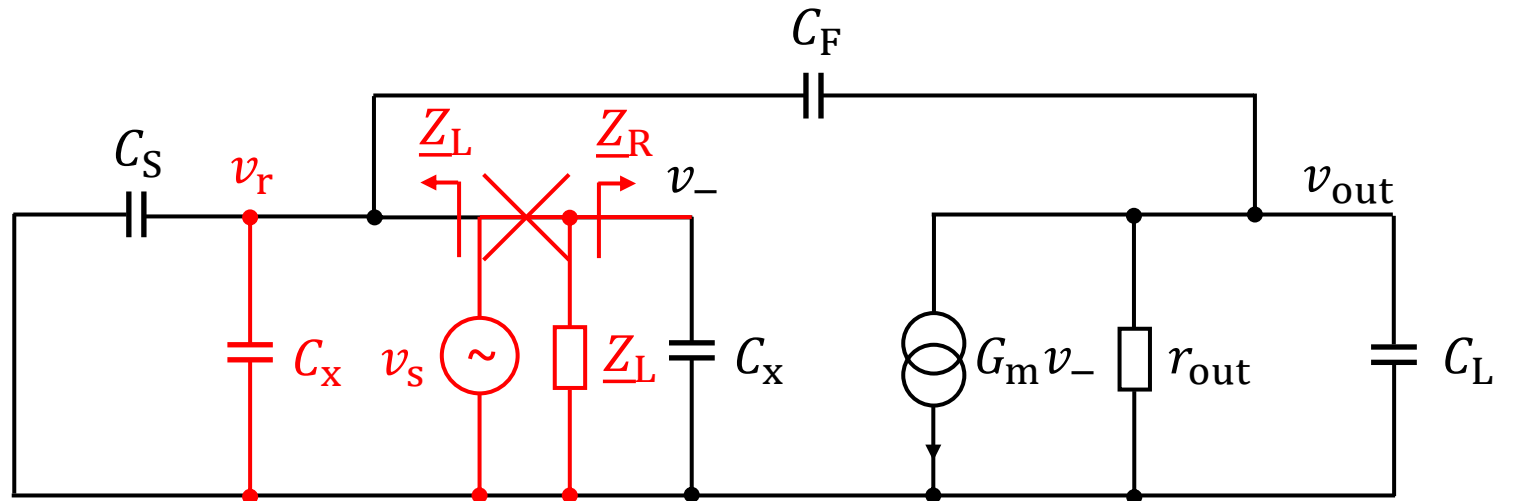
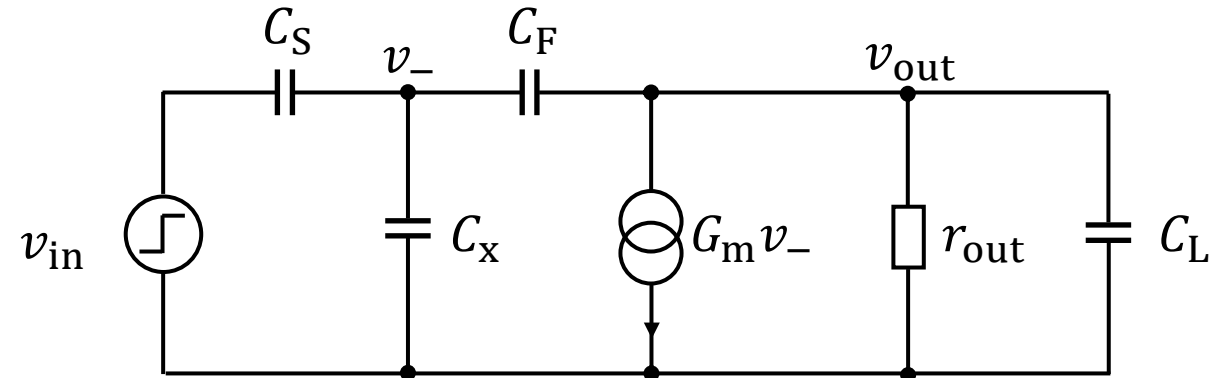
step III: determine $\underline{Z}_L$ and $\underline{Z}_R$

$$\underline{Z}_R = (sC_X)^{-1}$$

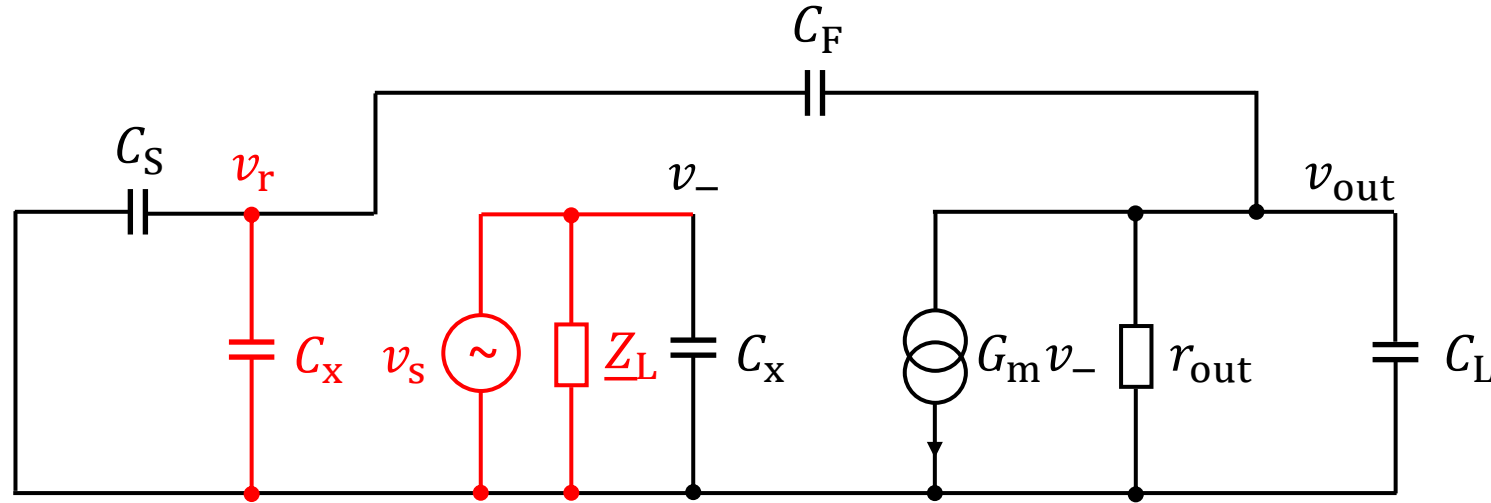
$\underline{Z}_L$ doesn't matter since...

step IV: ... it is driven by an ideal
signal source anyway

step VI: determine $L(s) = v_r/v_s$



TII.c – Determination Loop Gain $L(s)$



- G_m generates a current of $G_m v_s$ since $v_- = v_s$
- The current flows through the current divider $r_{out} || (C_F - C_S || C_X) || C_L$
- The voltage v_r equals the current in the branch $C_F - C_S || C_X$ multiplied by the impedance of $C_S || C_X$

$$L(s) = \frac{G_m C_F}{(C_F + C_S + C_X)g_{out} + (C_F C_L + C_F C_S + C_L C_S + C_F C_X + C_L C_X)s}$$

TII.d – Simplification for $g_{\text{out}} \gg C_i$

$$L(s) = \frac{G_m C_F}{(C_F + C_S + C_X)g_{\text{out}} + (C_F C_L + C_F C_S + C_L C_S + C_F C_X + C_L C_X)s}$$

g_{out} typically a few tens of μS $\leftrightarrow$ any C_i a few tens to hundreds of fF

→ It is okay to neglect the imaginary part of the denominator at low frequencies

since any $C_i C_j \ll C_k g_{\text{out}}$.

$$L(s) \approx \frac{G_m C_F}{(C_F + C_S + C_X)g_{\text{out}}} = \frac{C_F}{C_F + C_S + C_X} G_m r_{\text{out}}$$

TII.e – Comparison to $\beta \cdot G_m r_{out}$

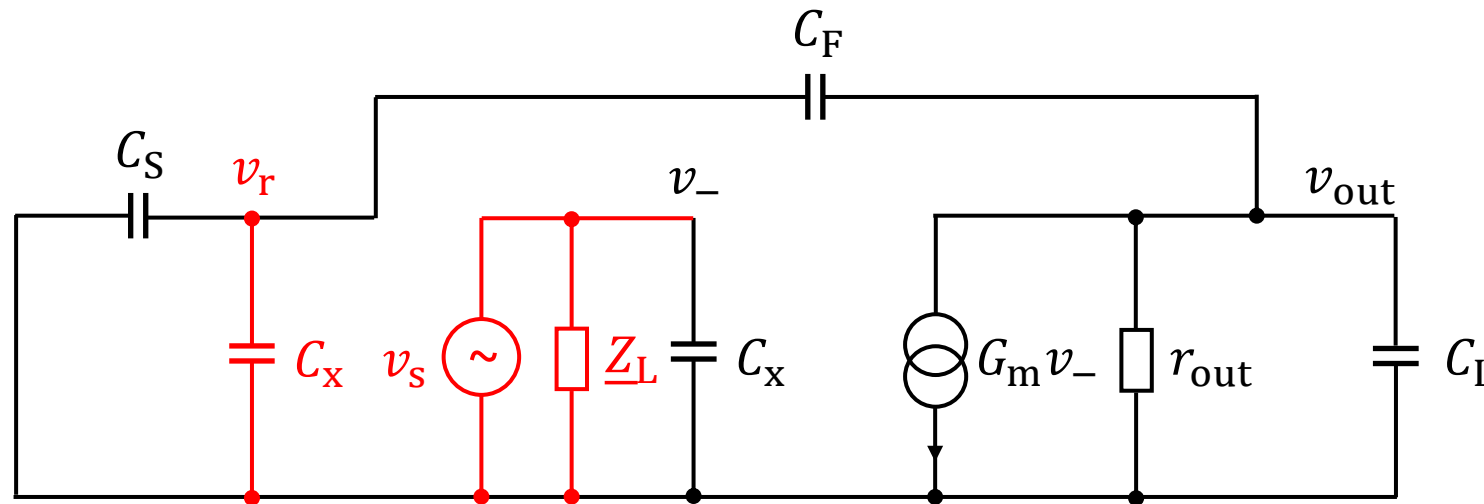
$$L(s) \approx \frac{C_F}{C_F + C_S + C_X} G_m r_{out}$$

return factor β

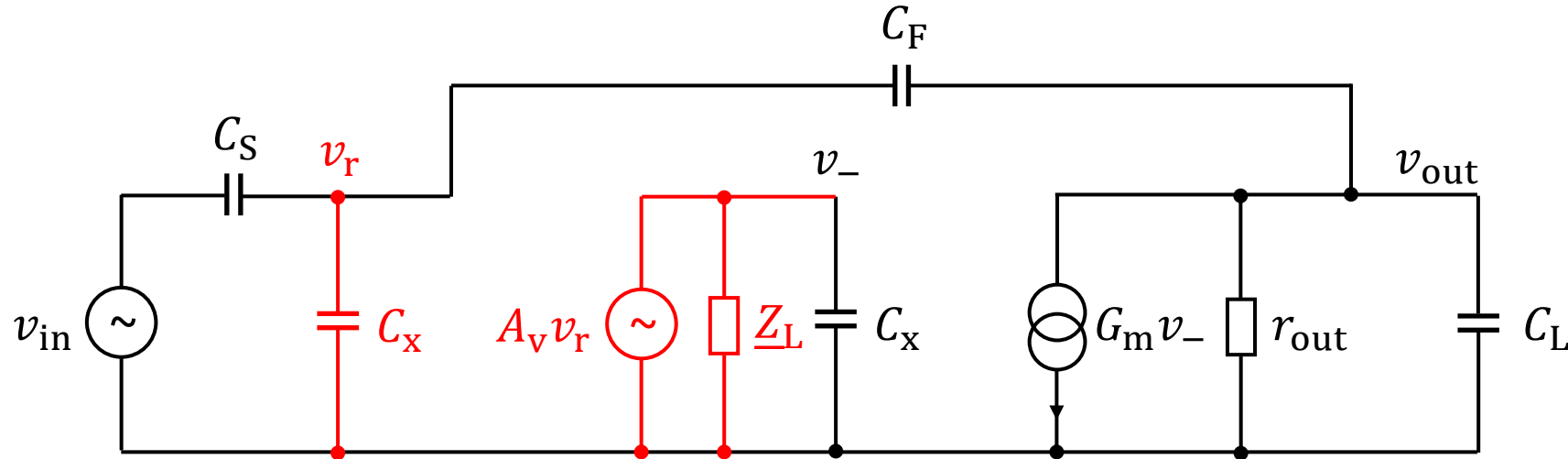
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comparison to $\beta \cdot G_m r_{out} \rightarrow \beta = \frac{v_r}{v_{out}} = \frac{C_F}{C_F + C_S + C_X}$

fraction of the output voltage v_{out}
that is fed back/returned to the input v_r
of the opamp.

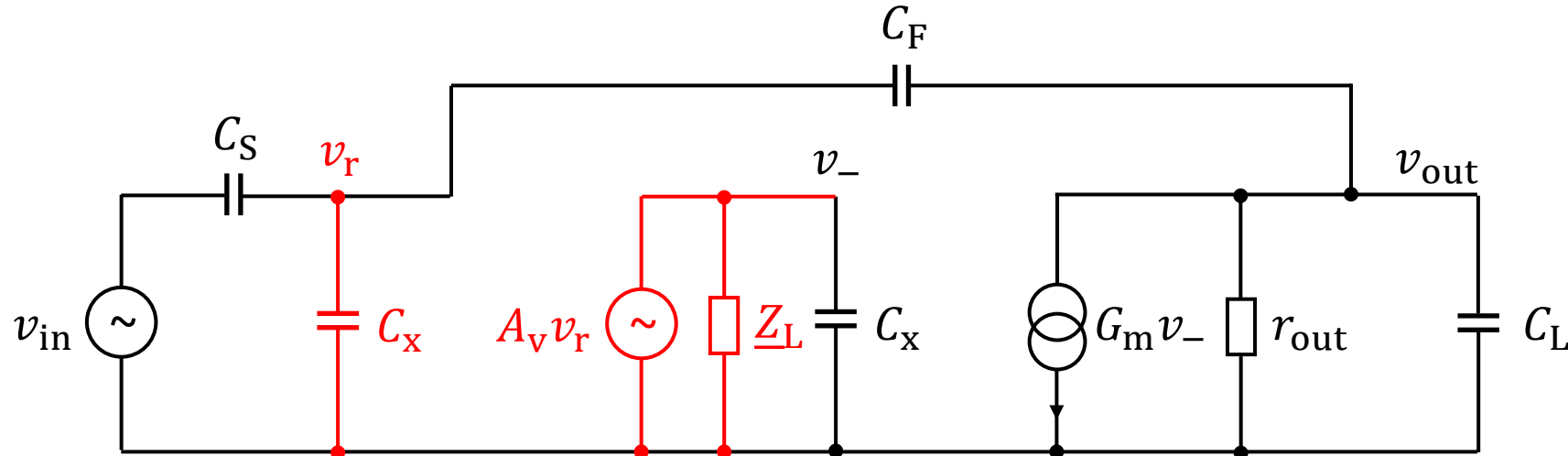


TII.f – Determination $A_{L\infty}(s)$



- Replace v_s by $A_V v_r$ and let $A_V \rightarrow \infty$ to determine $TF(s)_{L(s) \rightarrow \infty}$.
- However, since v_{out} is finite, it follows that $v_r \rightarrow 0 \text{ V}$ for $A_V \rightarrow \infty$.
 - If $v_r \rightarrow 0 \text{ V}$, it follows that $i_{C_X} = sC_X v_r \rightarrow 0 \text{ A}$.
- Since $i_{C_X} \rightarrow 0 \text{ A}$, it follows that $i_{C_S} = i_{C_F} \rightarrow A_{L\infty}(s) = \left. \frac{v_{out}}{v_{in}} \right|_{L(s) \rightarrow \infty} = -\frac{C_S}{C_F}$

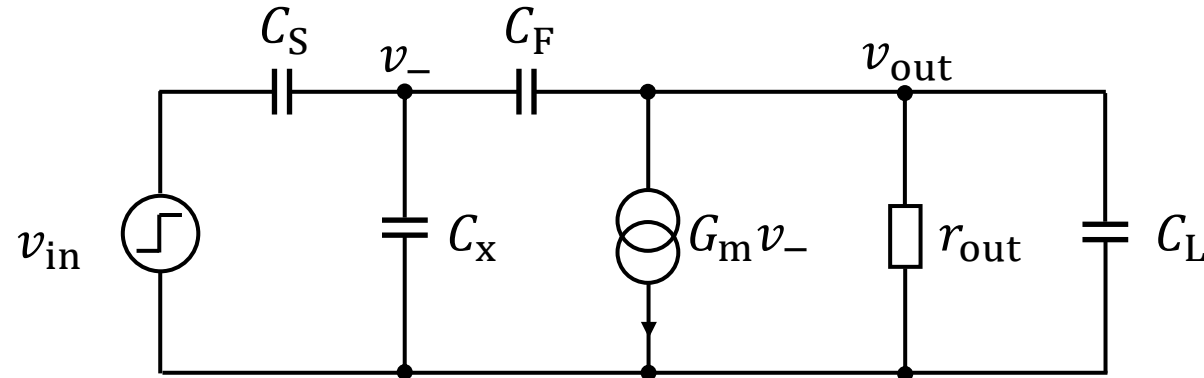
TII.g – Determination $A_{L0}(s)$



- Replace v_s by $A_V v_r$ and let $A_V \rightarrow 0$ to determine $TF(s)_{L(s) \rightarrow 0}$.
- However, since $v_- = A_V v_r = 0$ V, it follows that $G_m v_- = 0$ A for $A_V \rightarrow 0$.
- We thus get $TF(s)_{L(s) \rightarrow 0} = \frac{v_{out}}{v_{in}} \Big|_{L(s) \rightarrow 0} = \frac{v_{out}}{v_r} \frac{v_r}{v_{in}}$ (2x voltage dividers)

$$A_{L0}(s) = \frac{v_{out}}{v_{in}} \Big|_{L(s) \rightarrow 0} = \frac{C_S C_F s}{(C_F + C_S + C_X) g_{out} + (C_F C_L + C_F C_S + C_L C_S + C_F C_X + C_L C_X) s}$$

TII.h – Determination $TF_{AGF}(s)$



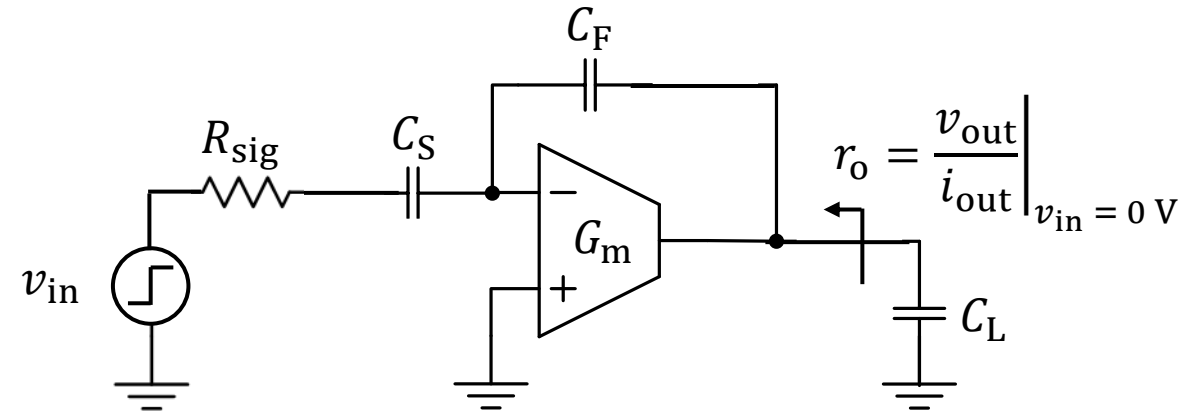
$$TF_{AGF}(s) = \frac{v_{out}}{v_{in}} = \frac{L(s)A_{L\infty}(s)}{1 + L(s)} + \frac{A_{L0}(s)}{1 + L(s)}$$

$$= \frac{-C_S(G_m - sC_F)}{(C_F + C_S + C_X)g_{out} + C_F G_m + s(C_F C_L + C_F C_S + C_L C_S + C_F C_X + C_L C_X)}$$

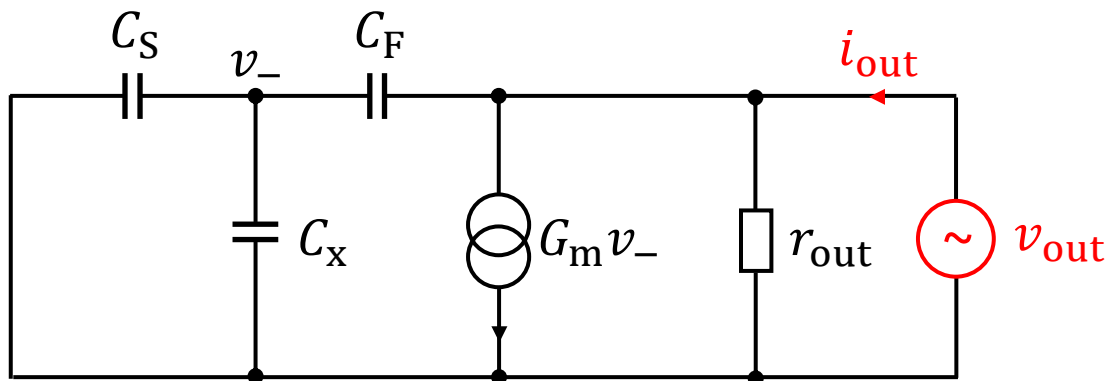
As expected, the result matches the result derived by the application of the nodal analysis in task I.

TII.i – Determination Output Resistance

Setting $v_{in} = 0$ V and neglecting the signal source resistance R_{sig} , we get



small-signal equivalent circuit

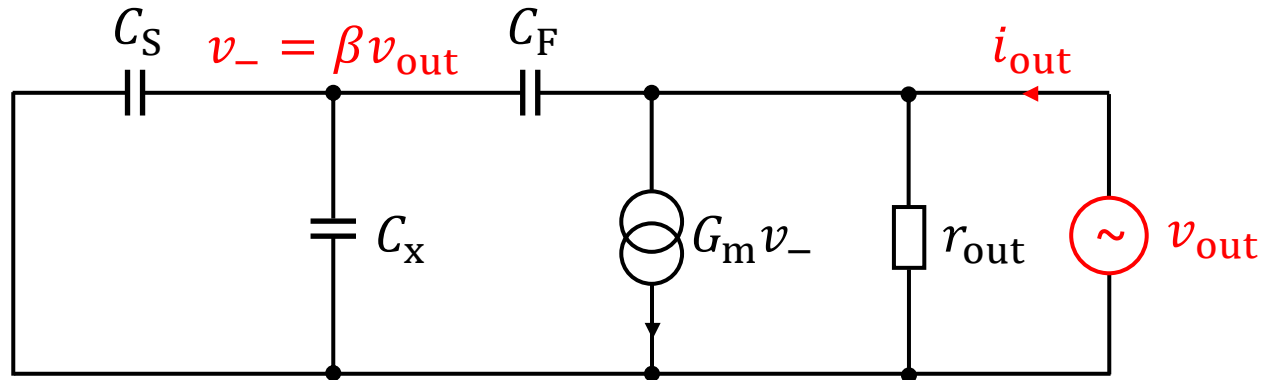


The resulting circuit is equivalent to the circuit used to determine $L(s)$.

$$\rightarrow v_- = \beta v_{out}$$

TII.i – Determination Output Resistance

small-signal equivalent circuit

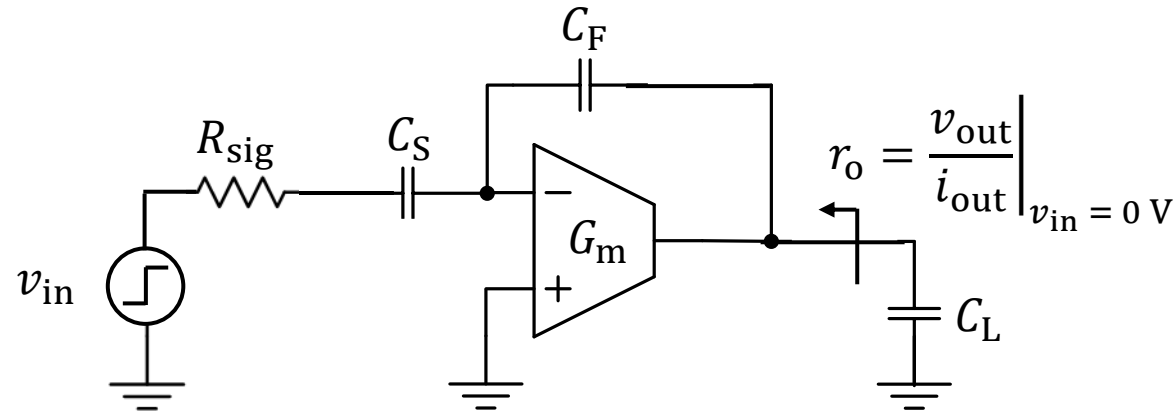


$$i_{\text{out}} = \frac{v_{\text{out}}}{r_{\text{out}}} + \beta G_m v_{\text{out}} + sC_F(1 - \beta)v_{\text{out}}$$

$$\approx \beta G_m v_{\text{out}} + sC_F(1 - \beta)v_{\text{out}}$$

$$\rightarrow r_o = \frac{v_{\text{out}}}{i_{\text{out}}} \approx \frac{1}{\beta G_m + s(1 - \beta)C_F} \equiv \frac{1}{\beta G_m} \parallel \frac{1}{s(1 - \beta)C_F}$$

TII.j – Determination Pole @ Output Node



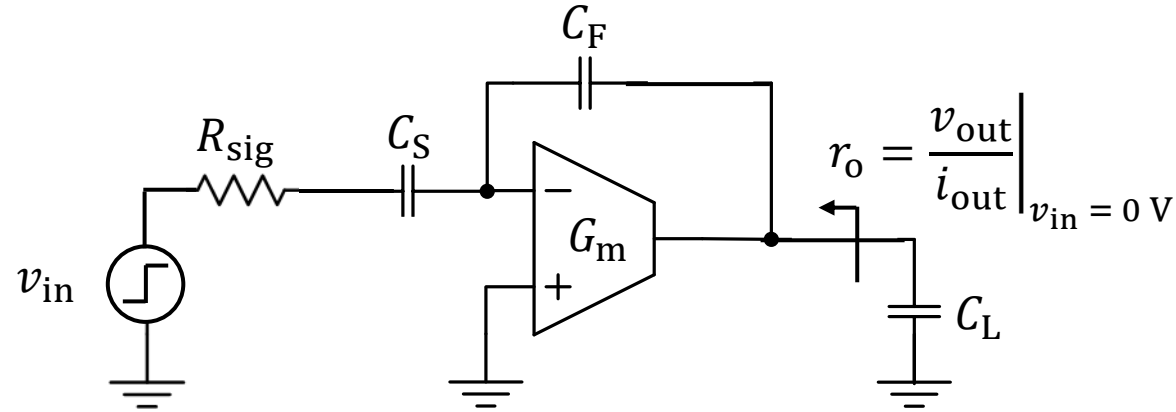
- Since C_L is connected parallel to r_o , the total impedance of the output node v_{out} results in

$$r_{total} \approx \frac{1}{\beta G_m} \parallel \frac{1}{s(1-\beta)C_F} \parallel \frac{1}{sC_L} = \frac{1}{\beta G_m} \parallel \frac{1}{s\{C_L + (1-\beta)C_F\}}$$

- Since a pole is defined by $1/RC$, the output pole thus results in

$$p \approx \frac{\beta G_m}{C_L + (1-\beta)C_F}$$

TII.j – Determination Pole @ Output Node



Replacing $\beta = \frac{C_F}{C_F + C_S + C_X}$ in $p \approx \frac{\beta G_m}{C_L + (1 - \beta)C_F}$ yields the expression derived in task I.g:

$$p = - \frac{G_m}{C_L + C_S + C_X + \frac{(C_X + C_S)C_L}{C_F}}$$

TIII.a – Static Settling Error

$$\begin{aligned}\varepsilon_s &= \frac{TF(s) - TF_{\text{ideal}}(s)}{TF_{\text{ideal}}(s)} = \frac{\frac{A_{L\infty}(s)L(s)}{1 + L(s)} - A_{L\infty}(s)}{A_{L\infty}(s)} \\ &= -\frac{1}{1 + L(s)} \approx -\frac{1}{L(s)}\end{aligned}$$

rule of thumb: $\varepsilon_s = -1\% \rightarrow L(s) = 100$

TIII.b – Dynamic Settling Error (no Slewing)

$$\begin{aligned}\epsilon_d &= \frac{TF(t) - TF_{ideal}(s)}{TF_{ideal}(s)} = \frac{-\frac{C_S}{C_F}(1 - e^{-t/\tau})v_{step} - \left(-\frac{C_S}{C_F}\right)v_{step}}{-\frac{C_S}{C_F}v_{step}} \\ &= -e^{-t/\tau}\end{aligned}$$

$$N = \frac{t}{\tau} = -\ln |\epsilon_d|$$

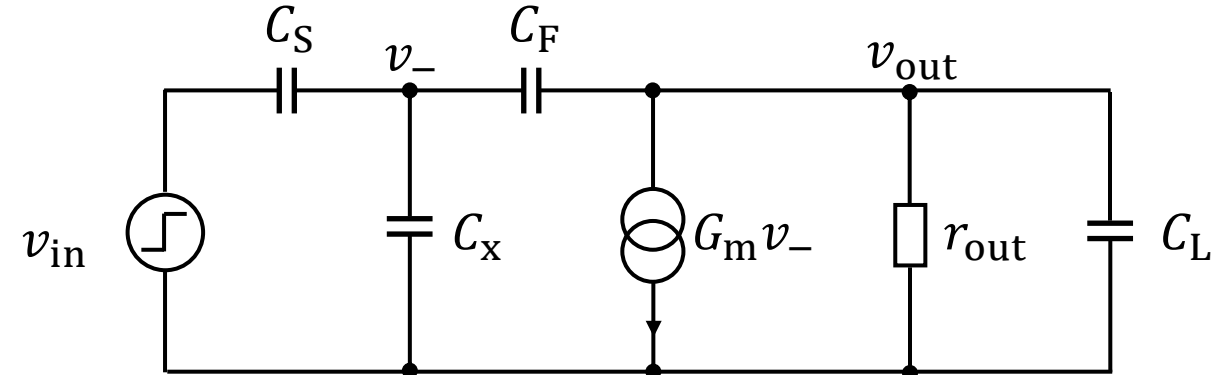
ϵ_d	N
-1.00%	4.6
-0.10%	6.9
-0.01%	9.2

TIV – Dynamic Settling Error (Slewing)

step response: $v_{\text{out}}(t) = -\frac{C_S}{C_F} (1 - e^{-\frac{t}{\tau}}) v_{\text{step}}$

$$= -(1 - e^{-\frac{t}{\tau}}) V_{\text{OUT}}$$

with $\tau \approx \frac{C_L + (1 - \beta)C_F}{\beta G_m} = \frac{C_{\text{Ltot}}}{\beta G_m}$



In order to charge C_{Ltot} accordingly, a current $i(t) \cong -C_{\text{Ltot}} \frac{\partial v_{\text{out}}(t)}{\partial t} = C_{\text{Ltot}} \frac{V_{\text{OUT}}}{\tau} e^{-t/\tau}$ is required.

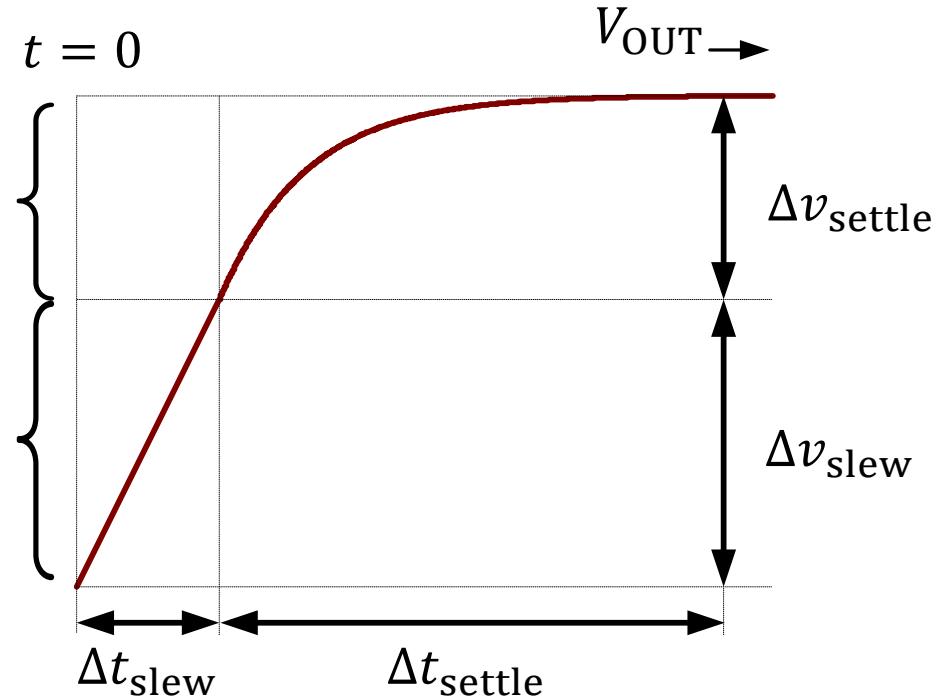
However, as long as $i(t)$ is larger than the maximum current I_{MAX} the amplifier can deliver, the output changes with a constant rate called slew rate SR , i.e.,

$$\left| \frac{\partial v_{\text{out}}(t)}{\partial t} \right| = \text{constant} = SR = \frac{I_{\text{MAX}}}{C_{\text{Ltot}}}$$

TIV – Dynamic Settling Error (Slewing)

$$v_{\text{out}}(t) = \Delta v_{\text{slew}} + \left(1 - e^{-\frac{t - \Delta t_{\text{slew}}}{\tau}}\right) \Delta v_{\text{settle}}$$

$$v_{\text{out}}(t) = SR \cdot t = \frac{I_{\text{MAX}}}{C_{\text{Ltot}}} t$$

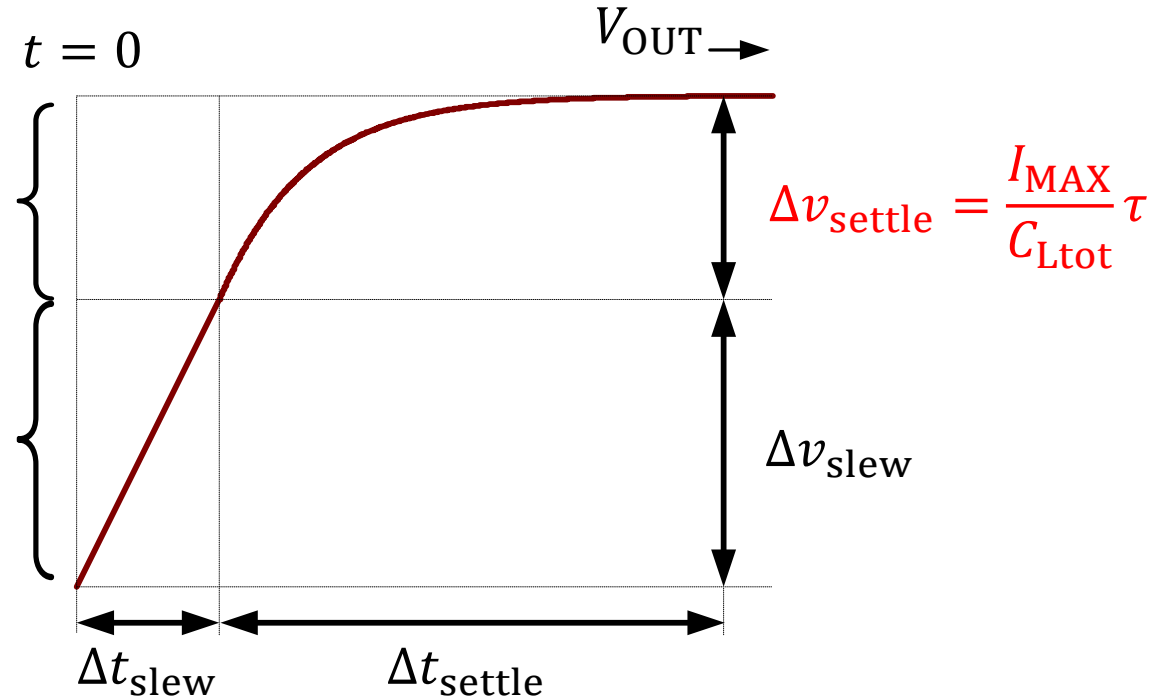


- Obviously, slewing affects the dynamic but not the static settling error.
- The derivative must be continuous in the transition point from slewing to settling.

TIV – Dynamic Settling Error (Slewing)

$$v_{\text{out}}(t) = \Delta v_{\text{slew}} + \left(1 - e^{-\frac{t - \Delta t_{\text{slew}}}{\tau}}\right) \Delta v_{\text{settle}}$$

$$v_{\text{out}}(t) = SR \cdot t = \frac{I_{\text{MAX}}}{C_{\text{Ltot}}} t$$



The derivative must be continuous in the transition point from slewing to settling.

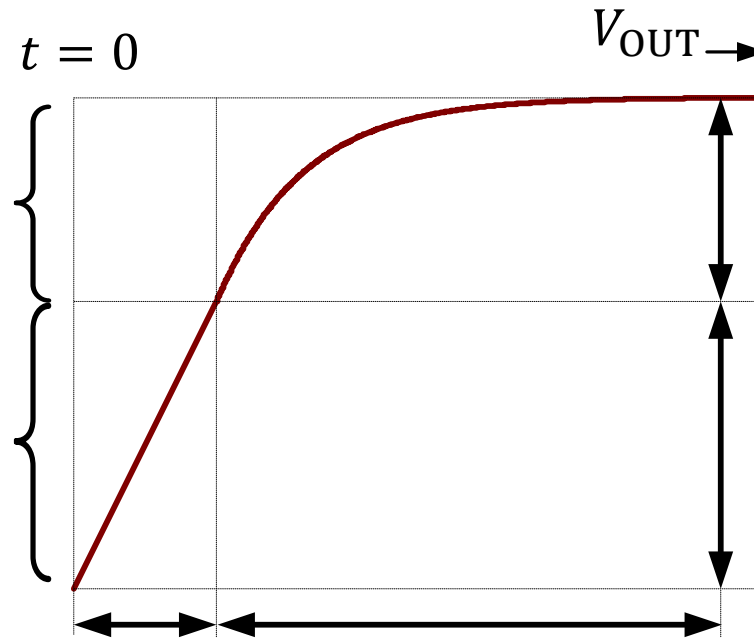
$$\left. \frac{I_{\text{MAX}}}{C_{\text{Ltot}}} = \frac{\Delta v_{\text{settle}}}{\tau} e^{-\frac{t - \Delta t_{\text{slew}}}{\tau}} \right|_{t=\Delta t_{\text{slew}}} = \frac{\Delta v_{\text{settle}}}{\tau} \rightarrow \Delta v_{\text{settle}} = \frac{I_{\text{MAX}}}{C_{\text{Ltot}}} \tau$$

TIV – Dynamic Settling Error (Slewing)

$$v_{\text{out}}(t) = \Delta v_{\text{slew}} + \left(1 - e^{-\frac{t - \Delta t_{\text{slew}}}{\tau}}\right) \Delta v_{\text{settle}}$$

$$v_{\text{out}}(t) = SR \cdot t = \frac{I_{\text{MAX}}}{C_{\text{Ltot}}} t$$

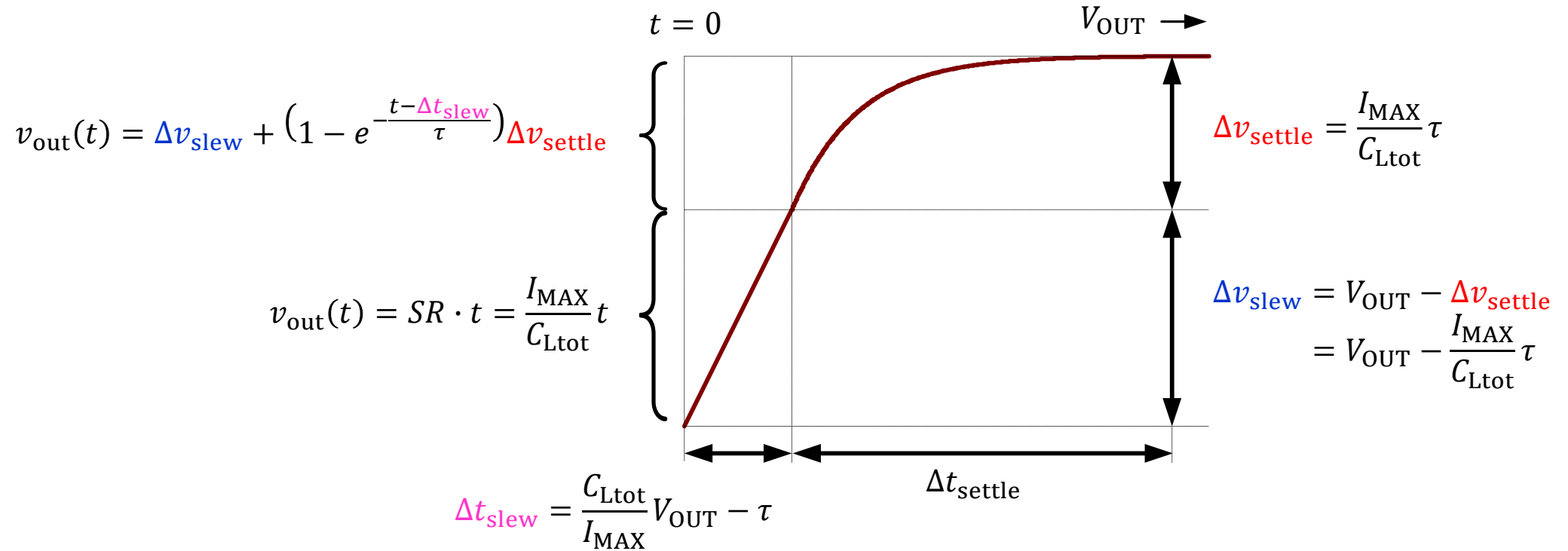
$$\begin{aligned} \Delta t_{\text{slew}} &= \frac{\Delta v_{\text{slew}}}{SR} \\ &= \frac{C_{\text{Ltot}}}{I_{\text{MAX}}} \Delta v_{\text{slew}} = \frac{C_{\text{Ltot}}}{I_{\text{MAX}}} V_{\text{OUT}} - \tau \end{aligned}$$



$$\Delta v_{\text{settle}} = \frac{I_{\text{MAX}}}{C_{\text{Ltot}}} \tau$$

$$\begin{aligned} \Delta v_{\text{slew}} &= V_{\text{OUT}} - \Delta v_{\text{settle}} \\ &= V_{\text{OUT}} - \frac{I_{\text{MAX}}}{C_{\text{Ltot}}} \tau \end{aligned}$$

TIV – Dynamic Settling Error (Slewing)



$$t > \Delta t_{slew}: \varepsilon_d = \frac{v_{out}(t) - V_{OUT}}{V_{OUT}} = \frac{\Delta v_{slew} + \left(1 - e^{-\frac{t - \Delta t_{slew}}{\tau}}\right) \Delta v_{settle} - V_{OUT}}{V_{OUT}} = -\frac{e^{-\frac{t - \Delta t_{slew}}{\tau}} \Delta v_{settle}}{V_{OUT}}$$

TIV – Dynamic Settling Error (Slewing)

$$t > \Delta t_{\text{slew}}: \varepsilon_d = \frac{v_{\text{out}}(t) - V_{\text{OUT}}}{V_{\text{OUT}}} = \frac{e^{-\frac{t - \Delta t_{\text{slew}}}{\tau}} \Delta v_{\text{settle}}}{V_{\text{OUT}}}$$

$$V_{\text{OUT}} = \frac{C_S}{C_F} v_{\text{step}}$$

$$\tau \approx \frac{C_L + (1 - \beta)C_F}{\beta G_m} = \frac{C_{\text{Ltot}}}{\beta G_m}$$

$$v_{\text{out}}(t) = \Delta v_{\text{slew}} + \left(1 - e^{-\frac{t - \Delta t_{\text{slew}}}{\tau}}\right) \Delta v_{\text{settle}}$$

$$v_{\text{out}}(t) = SR \cdot t = \frac{I_{\text{MAX}}}{C_{\text{Ltot}}} t$$

$$\Delta t_{\text{slew}} = \frac{C_{\text{Ltot}}}{I_{\text{MAX}}} V_{\text{OUT}} - \tau$$

